APPENDIX A
User guide

A.1 Purpose and requirements. The application «Sea Course GA» finds the course of a sea vessel between two positions over the water points of a labelled dataset with a genetic algorithm and compares it with the optimal routes of Dijkstra's and A* algorithms. It requires Windows 10/11 or Linux (Ubuntu 20.04+, Fedora 38+) and Python 3.10 or newer; on Ubuntu 20.04, where the system Python is 3.8, Python 3.10 has to be installed additionally. A computer with at least 4 GB of RAM is recommended: reading the dataset of the project needs about 2 GB of memory at its peak.

A.2 Installation. The folder of the project is copied to the computer, and the dependencies are installed into a virtual environment. Under Windows:

```
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements-course.txt
```

Under Linux:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements-course.txt
```

If Qt reports on Ubuntu that the platform plugin «xcb» cannot be loaded, the package `libxcb-xinerama0` has to be installed. The dataset is placed in the folder `data/raw` under the name `merged.json`.

A.3 Start. The application is started from the folder of the project:

```
python scripts/run_desktop.py
python scripts/run_desktop.py --dataset path/to/points.geojson
```

Without the option `--dataset` the file `data/raw/merged.json` is opened automatically if it exists. Another file is opened with the menu «File → Open dataset...» (Ctrl+O) or the button «Open...». The file must contain the latitude, the longitude and the attribute `class_code` with one of the values `OPEN_SEA`, `COASTAL_SEA`, `NEAR_COAST`, `COASTLINE`.

A.4 Work with the application. A typical session consists of the following steps.

1) Check in the group «1. Dataset» that the file is loaded and the number of points of every class is shown. The check box «Remove class/depth conflicts (5 m)» should normally stay on.

2) In the group «2. Route» choose a preset voyage or type the coordinates of the departure and the destination in decimal degrees. The positions can also be picked on the map: press «Pick departure» or «Pick destination» and click on the map.

3) In the group «3. Graph» keep the area «Around the route» and the edge mode «By visibility». Press «Build graph». The check box «Show edges» draws the edges on the map (figure A.1).

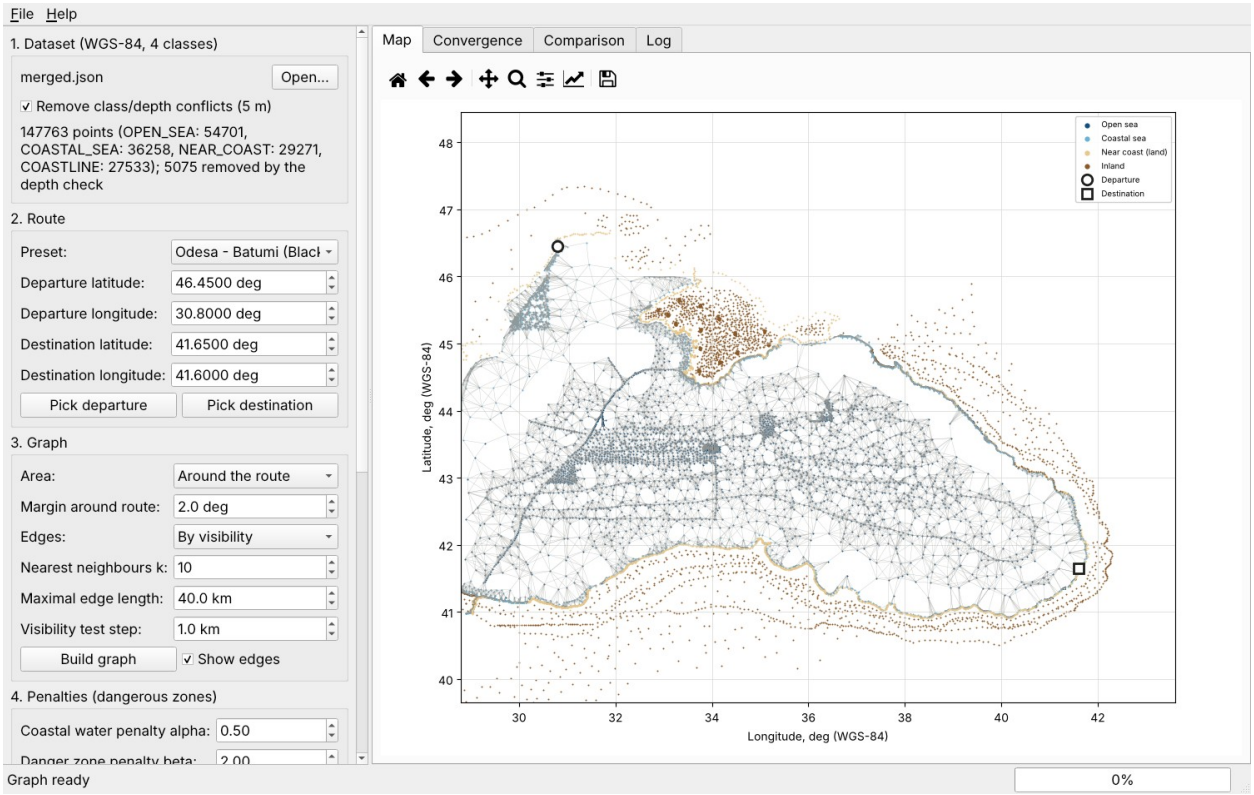


Figure A.1 – Main window after building the graph

4) In the group «4. Penalties» set the coastal water penalty α, the danger zone penalty β and the danger zone radius r. A change of these values recomputes the edge costs and clears the previous results.

5) In the group «5. Genetic algorithm» set the parameters of the algorithm (figure A.2). The same random seed gives the same route.

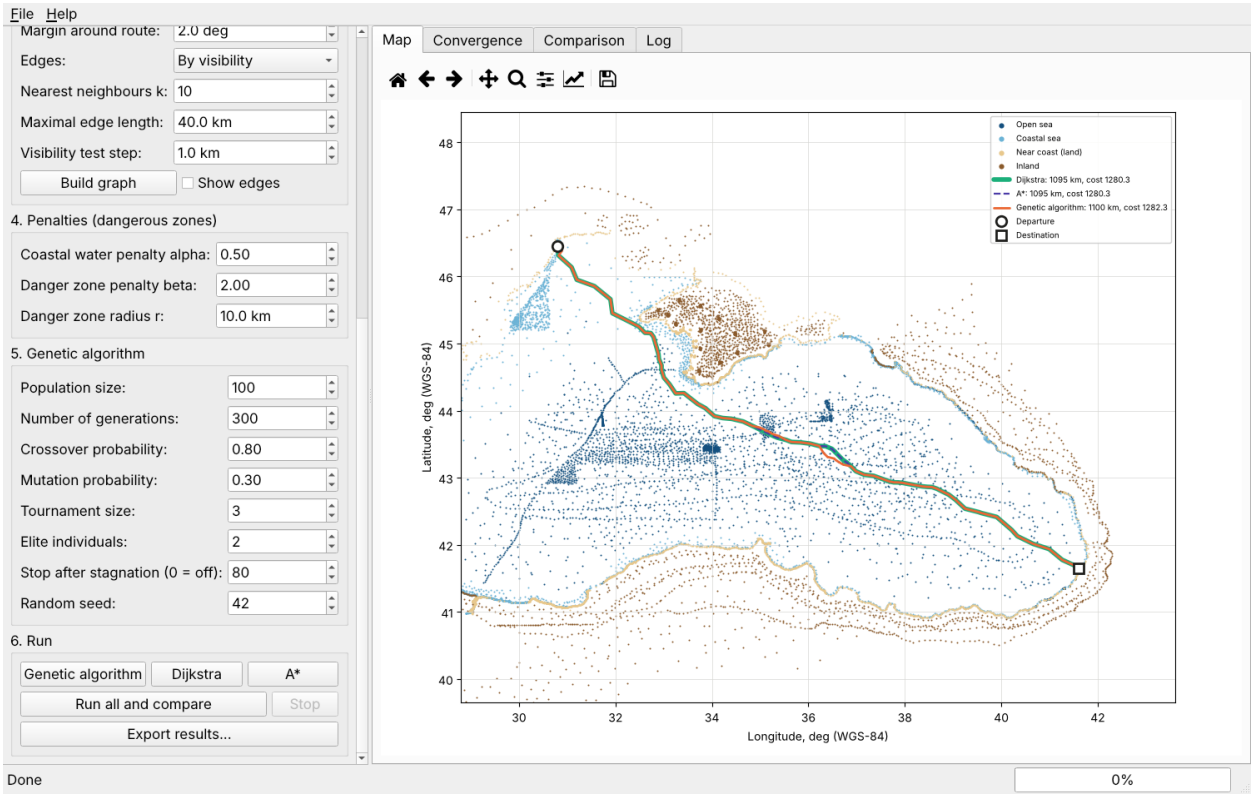


Figure A.2 – Groups «Penalties», «Genetic algorithm» and «Run» of the control panel

6) Press «Run all and compare» or one of the buttons «Genetic algorithm», «Dijkstra», «A*». The status bar shows the current generation and the best cost; «Stop» ends the genetic algorithm after the current generation. If the graph is out of date, it is rebuilt first.

7) Read the results on the tabs «Map» (figure A.3), «Convergence» (figure A.4) and «Comparison» (figure A.5). The column «Gap, %» shows the excess of the cost over the optimum of Dijkstra/A*.

8) Press «Export results...» (Ctrl+E) and choose a folder; the files are listed in table A.1.

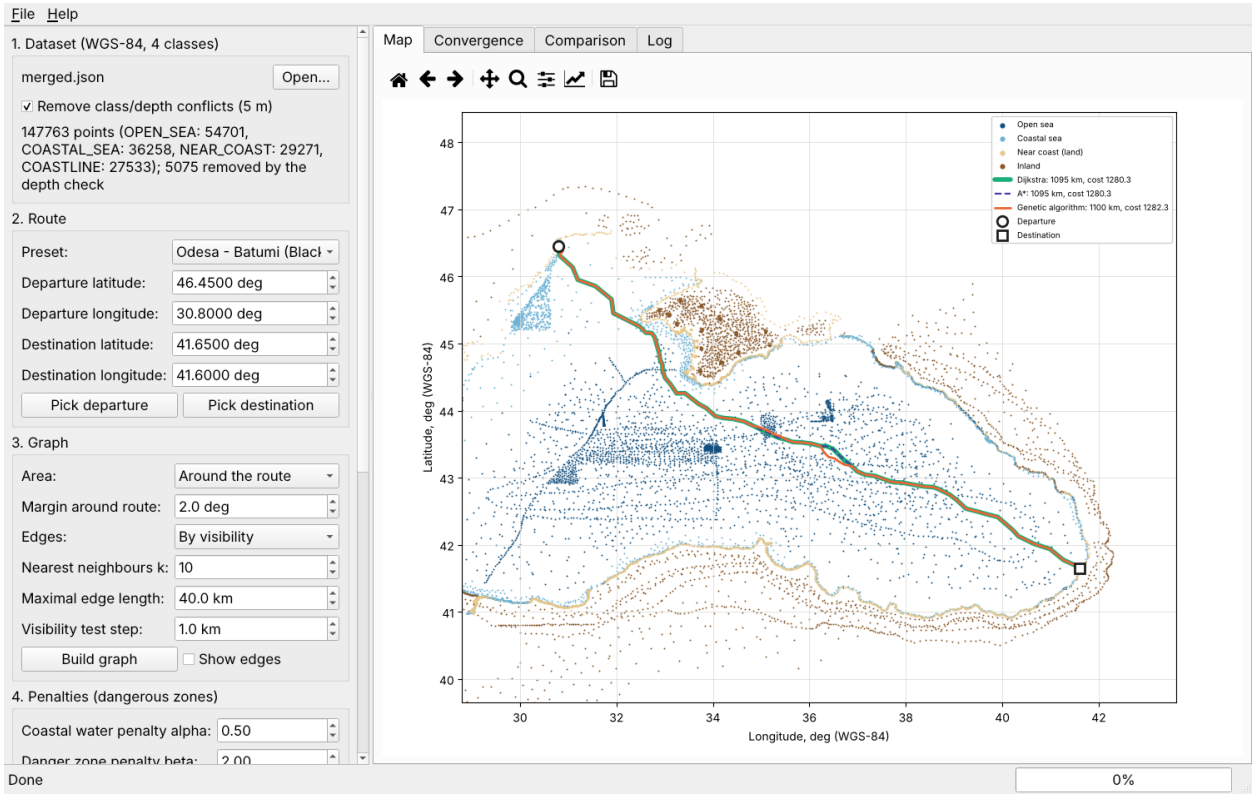


Figure A.3 – Routes of the voyage Odesa – Batumi on the tab «Map»

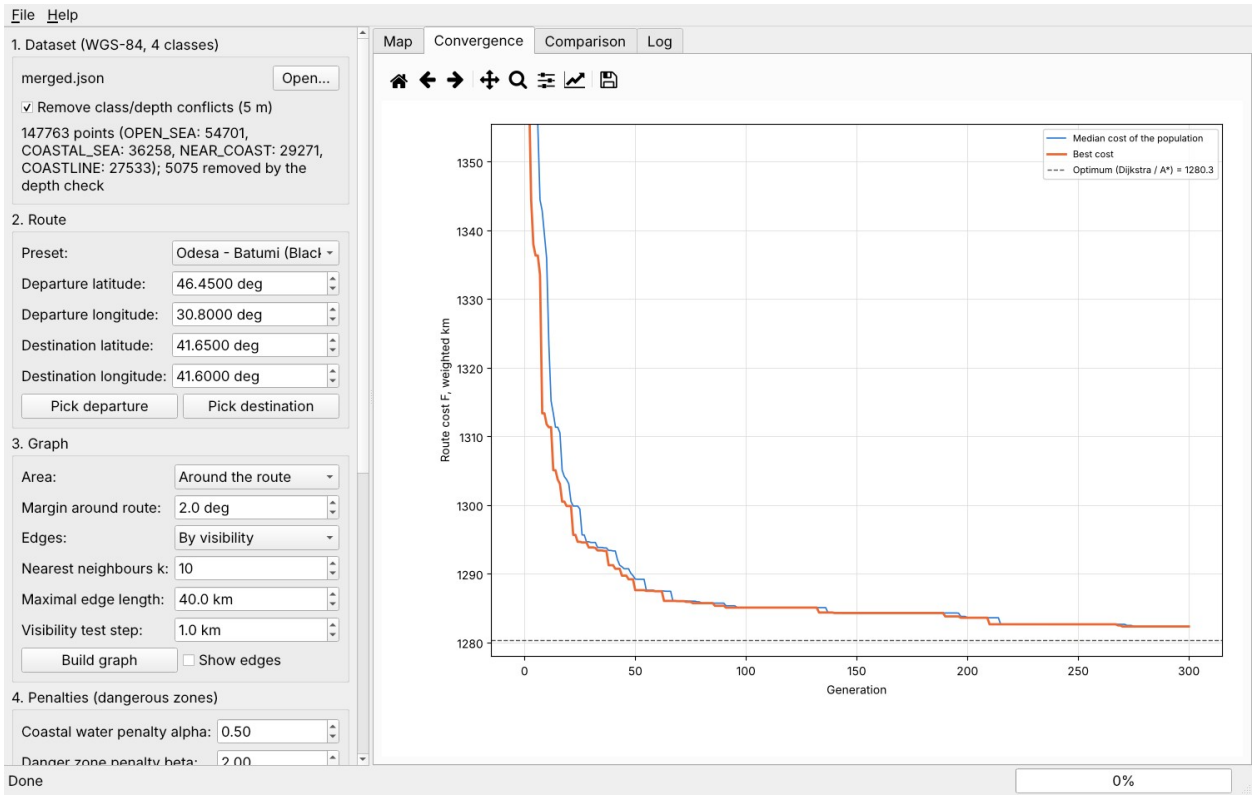


Figure A.4 – Tab «Convergence»

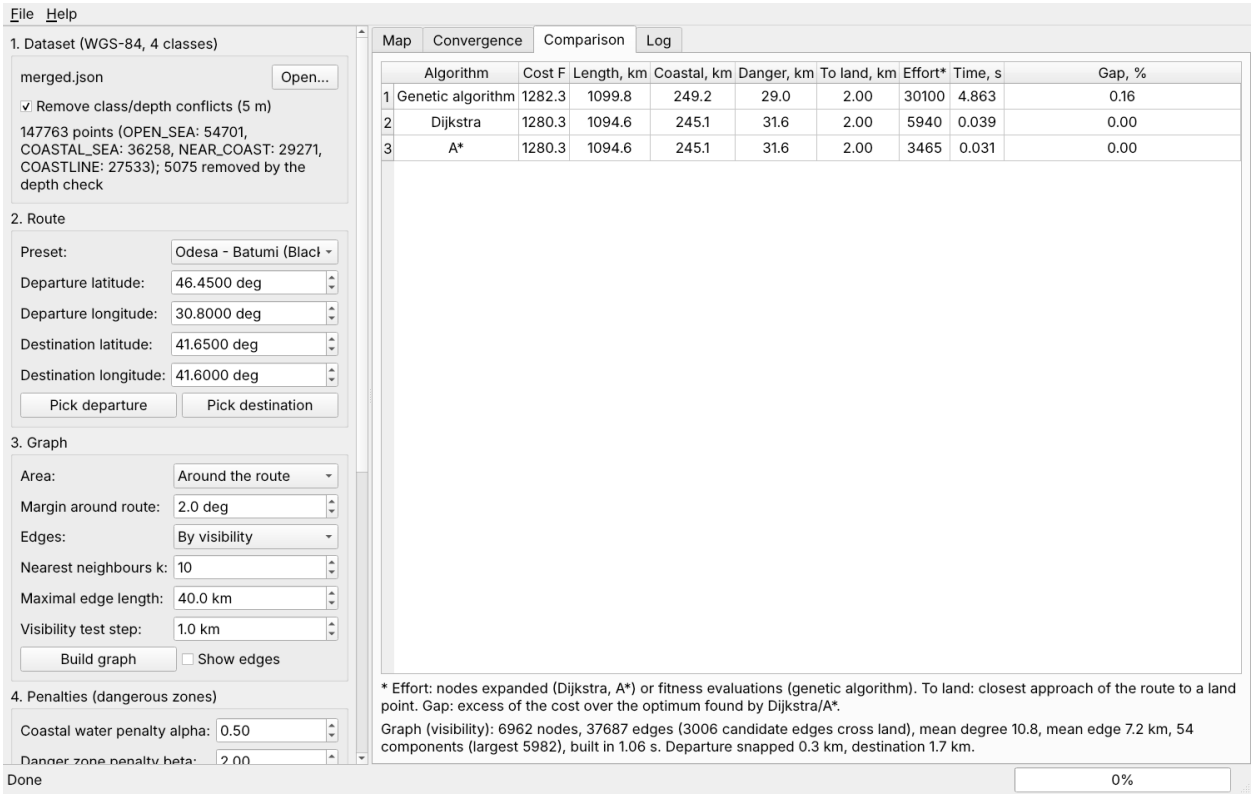


Figure A.5 – Tab «Comparison»

Table A.1 – Files written by the export (prefix «course_<date>_<time>»)

File	Content
`<prefix>_map.png`	map with the points and the routes
`<prefix>_convergence.png`	convergence plot of the genetic algorithm
`<prefix>_routes.geojson`	routes as GeoJSON lines, departure and destination as points
`<prefix>_routes.csv`	waypoints of every route: algorithm, number, latitude, longitude
`<prefix>_comparison.csv`	comparison table of the algorithms
`<prefix>_log.json`	all parameters and results in machine-readable form
`<prefix>_log.txt`	the same in readable form
`<prefix>_map.html`	interactive map (only if folium is installed)

A.5 Messages. The messages of the application and the actions of the user are listed in table A.2. The full text of every error, including the technical details, is written to the tab «Log».

Table A.2 – Messages of the application

Message	Cause and action
«Open the dataset first.»	a graph or a search was requested before loading a file; open a dataset
«The file has no 'class_code' attribute – a labelled dataset is required»	the file has no classes; use a labelled dataset
«No labelled points remain after validation»	no valid point in the file; check the coordinates and the class codes
«The selected area contains fewer than two water points»	the area has no water; move the end points or increase the margin
«No path: start and goal lie in different components»	the end points cannot be joined on this graph; increase the margin or the maximal edge length
«Warning: an end point is far from the nearest graph node»	the dataset has few water points near the end point; the route starts or ends at the nearest node
«Run at least one algorithm first.»	export without results; run an algorithm